Mechanical Engineering Experiments — Report 1

機械工程實驗(一) 第一次報告

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Q1:

(a) . By your collected data in Table 1. Estimate and report the value: $V_{average}^2 + V_{AC,mutimeter}^2$ for all four rows.

A. Sinusoidal with 1V offset:

$V_{average}^2 + V_{AC,mutimeter}^2 = 1.020^2 + 1.126^2 \approx 2.308$

B. Sinusoidal without offset:

$V_{average}^2 + V_{AC,mutimeter}^2 = 0.340^2 + 1.110^2 \approx 1.348$

C. Square with 1V offset:

$V_{average}^2 + V_{AC,mutimeter}^2 = 0.996^2 + 1.503^2 \approx 3.251$

D. Square without offset:

$V_{average}^2 + V_{AC,mutimeter}^2 = 0.370^2 + 1.492^2 \approx 2.363$

(b) . If we define the results from Q1(a) as $V_{RMS,measure}^2$, estimate and report the error between $V_{RMS,measure}^2$ and V_{RMS}^2 of Table 1. The error could be estimated follows the equation: $Measurement\ error = \frac{V_{RMS,measure}^2 - V_{RMS}^2}{V_{RMS}^2} \times 100\%$

A. Sinusoidal with 1V offset:

$V_{RMS} = 1.52\text{V}$

$V_{RMS}^2 = 1.52^2 \approx 2.310$

$V_{RMS,measure}^2 = 2.308$

Measurement error $= \frac{2.308 - 2.310}{2.310} \times 100\% \approx -0.0866\%$

B. Sinusoidal without offset:

$V_{RMS} = 1.17\text{V}$

$V_{RMS}^2 = 1.17^2 \approx 1.369$

$V_{RMS,measure}^2 = 1.348$

Measurement error $= \frac{1.348 - 1.369}{1.369} \times 100\% \approx -1.534\%$

C. Square with 1V offset:

$V_{RMS} = 1.90\text{V}$

$V_{RMS}^2 = 1.90^2 = 3.610$

$V_{RMS,measure}^2 = 3.251$

Measurement error = $\frac{3.251-3.610}{3.610} \times 100\% \approx$ -9.945%

D. Square without offset:

$V_{RMS} = 1.67\text{V}$

$V_{RMS}^2 = 1.67^2 \approx 2.789$

$V_{RMS,measure}^2 = 2.363$

Measurement error = $\frac{2.363-2.789}{2.789} \times 100\% \approx$ -15.274%

(c) . Repeat Q1(a) and Q1(b) but replace $V_{average}^2$ by $V_{DC,multimeter}^2$. Which instrument (oscilloscope vs. digital meter) provides a better approach with respect to V_{RMS}^2?

A. Sinusoidal with 1V offset:

$V_{RMS,measure}^2 = V_{DC,mutimeter}^2 + V_{AC,mutimeter}^2 = 1.013^2 + 1.126^2 \approx 2.294$

$V_{RMS}^2 = 1.52^2 \approx 2.310$

Measurement error = $\frac{2.294-2.310}{2.310} \times 100\% \approx$ -0.693%

B. Sinusoidal without offset:

$V_{RMS,measure}^2 = V_{DC,mutimeter}^2 + V_{AC,mutimeter}^2 = 0.349^2 + 1.110^2 \approx 1.354$

$V_{RMS}^2 = 1.17^2 \approx 1.369$

Measurement error = $\frac{1.354-1.369}{1.369} \times 100\% \approx$ -1.096%

C. Square with 1V offset:

$V_{RMS,measure}^2 = V_{DC,mutimeter}^2 + V_{AC,mutimeter}^2 = 0.979^2 + 1.503^2 \approx 3.217$

$V_{RMS}^2 = 1.90^2 = 3.610$

Measurement error = $\frac{3.217-3.610}{3.610} \times 100\% \approx$ -10.886%

D. Square without offset:

$V_{RMS,measure}^2 = V_{DC,mutimeter}^2 + V_{AC,mutimeter}^2 = 0.363^2 + 1.492^2 \approx 2.358$

$V_{RMS}^2 = 1.67^2 \approx 2.789$

Measurement error = $\frac{2.358-2.789}{2.789} \times 100\% \approx$ -15.454%

➔ The measurement error using $V_{DC,mutimeter}^2 + V_{AC,mutimeter}^2$ is larger than $V_{average}^2 + V_{AC,mutimeter}^2$, showing that oscilloscope have more accuracy.

(d) . Explain the physical meanings of crest factor clearly. And calculate all crest factors from Table 1. Does the experimental result match with your explanation?

Crest factor shows how "peaky" a waveform is compared to its average power level.

$$Crest\ Factor = \frac{V_{max}}{V_{RMS}}$$

If a waveform has a large crest factor value, it will show a sharper configuration (ex. Pulse train).

In contrast, if a waveform has a small crest factor value, it will show a flatter configuration (ex. Square wave).

A. Sinusoidal with 1V offset:

V_{max} = 2.68V

V_{RMS} = 1.52V

$$Crest\ Factor = \frac{V_{max}}{V_{RMS}} = \frac{2.68}{1.52} \approx 1.763$$

B. Sinusoidal without offset:

V_{max} = 2.00V

V_{RMS} = 1.17V

$$Crest\ Factor = \frac{V_{max}}{V_{RMS}} = \frac{2.00}{1.17} \approx 1.709$$

C. Square with 1V offset:

V_{max} = 2.68V

V_{RMS} = 1.90V

$$Crest\ Factor = \frac{V_{max}}{V_{RMS}} = \frac{2.68}{1.90} \approx 1.411$$

D. Square without offset:

V_{max} = 2.03V

V_{RMS} = 1.67V

$$Crest\ Factor = \frac{V_{max}}{V_{RMS}} = \frac{2.03}{1.67} \approx 1.216$$

➔ The experimental results match with the explanation of crest factor.

Sinusoidal waveforms have a larger crest factor value and square waveforms have a smaller crest factor value in contrast.

(e) . Explain why the rise time changes with different TIMEDIV scales?

Larger TIMEDIV scale (Oscilloscope reduces its sampling rate. Each sample point represents a longer time interval): The rising edge takes up only a small portion of the screen because the display covers a wider time span, making the rise time appear to be shorter and compressed.

Smaller TIMEDIV scale (Oscilloscope increases its sampling rate. More points are captured along the rising edge): The rising edge is expanded on the screen because the display covers a narrower time span, making the rise time appear to be longer and stretched.

Note: The "true rise time" of the signal does not change due to TIMEDIV scales. Only the oscilloscope's display and measurement resolution are affected.

Q2:

(a) . By your collected data in Table 2. Plot the dB-frequency curve for inverting circuit and low pass filtered circuit, respectively.

$$dB = 20\log\left(\frac{V_{pp,CH2}}{V_{pp,CH1}}\right)$$

A. Inverting circuit

Table 1. dB-Frequency Chart

Frequency(Hz)	Vpp,CH2	Vpp,CH1	Vpp,CH2 / Vpp,CH1	20log(Vpp,CH2 / Vpp,CH1)
50	10.1	2.01	5.024875622	14.02250633
99.9	10	2.01	4.975124378	13.93607885
200	10.01	2.01	4.980099502	13.9447604
5032	10.01	2.07	4.835748792	13.68927464
10010	10	2.07	4.830917874	13.68059309
14990	10	2.07	4.830917874	13.68059309
20000	9.76	2.09	4.669856459	13.38607063
29970	7.59	2.16	3.513888889	10.91576049
40050	5.92	2.2	2.690909091	8.597980518
44960	5.15	2.2	2.340909091	7.387690964
50000	4.63	2.22	2.085585586	6.384560331
55020	4.28	2.22	1.927927928	5.701815891
59940	3.92	2.22	1.765765766	4.938661851
64940	3.64	2.22	1.63963964	4.294968184
69940	3.31	2.22	1.490990991	3.469500387
75180	3.11	2.22	1.400900901	2.928148292
80070	2.96	2.22	1.333333333	2.498774732
85120	2.75	2.22	1.238738739	1.859594388
89960	2.59	2.22	1.166666667	1.338935793
95010	2.48	2.22	1.117117117	0.961974128
96240	2.4	2.22	1.081081081	0.677165345
97040	2.38	2.22	1.072072072	0.604479652
97990	2.35	2.22	1.058558559	0.494297756
99000	2.35	2.22	1.058558559	0.494297756
100000	2.31	2.22	1.040540541	0.345180109
101100	2.29	2.22	1.031531532	0.269650158
102000	2.25	2.22	1.013513514	0.116590873
103100	2.24	2.22	1.009009009	0.077900878
104000	2.24	2.22	1.009009009	0.077900878
104900	2.22	2.22	1	0

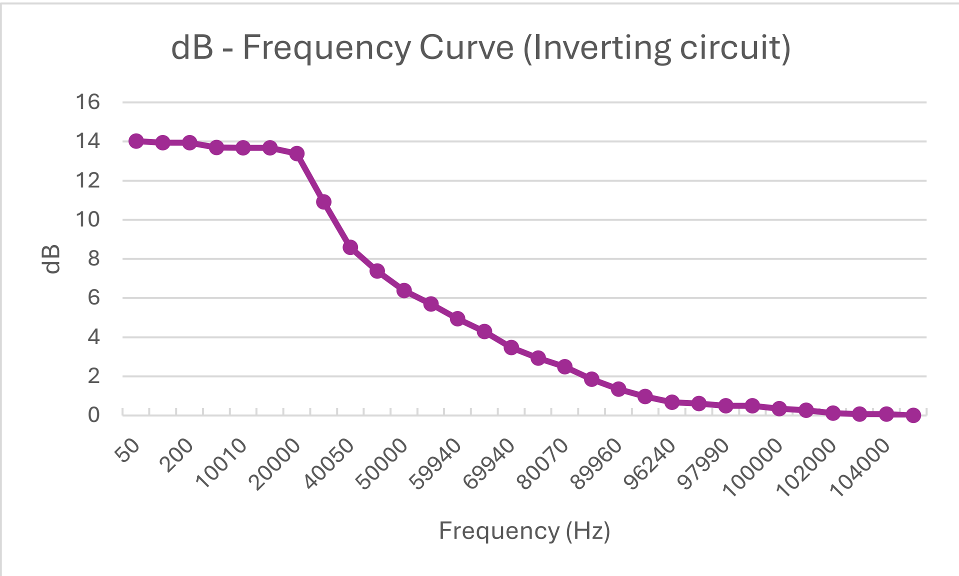

Figure 1. dB-Frequency Curve

B. Low pass filter

Table 2. dB-Frequency Chart (Low pass filter)

Frequency(Hz)	Vpp,CH2	Vpp,CH1	Vpp,CH2 / Vpp,CH1	20log(Vpp,CH2 / Vpp,CH1)
50	8.31	2.05	4.053658537	12.15694325
150.2	8	2.05	3.902439024	11.82672252
250.6	7.59	2.05	3.702439024	11.3697583
349.2	7.19	2.05	3.507317073	10.89950059
451.4	6.71	2.05	3.273170732	10.29937318
550	6.15	2.05	3	9.542425094
651.6	6.76	2.05	3.297560976	10.3638567
750.7	5.19	2.05	2.531707317	8.068269936
850.5	4.8	2.05	2.341463415	7.389747526
950.8	4.4	2.05	2.146341463	6.633976309
1046	4.23	2.05	2.063414634	6.291730126
1149	4	2.05	1.951219512	5.806122605
1251	3.75	2.05	1.829268293	5.245548133
1349	3.51	2.05	1.712195122	4.671065108
1400	3.44	2.07	1.661835749	4.411761942
1450	3.35	2.07	1.618357488	4.181489232
1499	3.2	2.07	1.54589372	3.783592657
1550	3.11	2.07	1.502415459	3.535800871
1599	3.11	2.09	1.488038278	3.452282058
1700	2.96	2.09	1.416267943	3.022908499
1801	2.79	2.09	1.33492823	2.509158343
1899	2.64	2.09	1.263157895	2.029152815
2002	2.64	2.09	1.263157895	2.029152815
2101	2.4	2.09	1.148325359	1.201299112
2202	2.31	2.09	1.105263158	0.869313876
2304	2.16	2.09	1.033492823	0.286149301
2319	2.16	2.11	1.023696682	0.203425917
2362	2.14	2.11	1.014218009	0.122626361
2380	2.11	2.11	1	0
2401	2.09	2.09	1	0

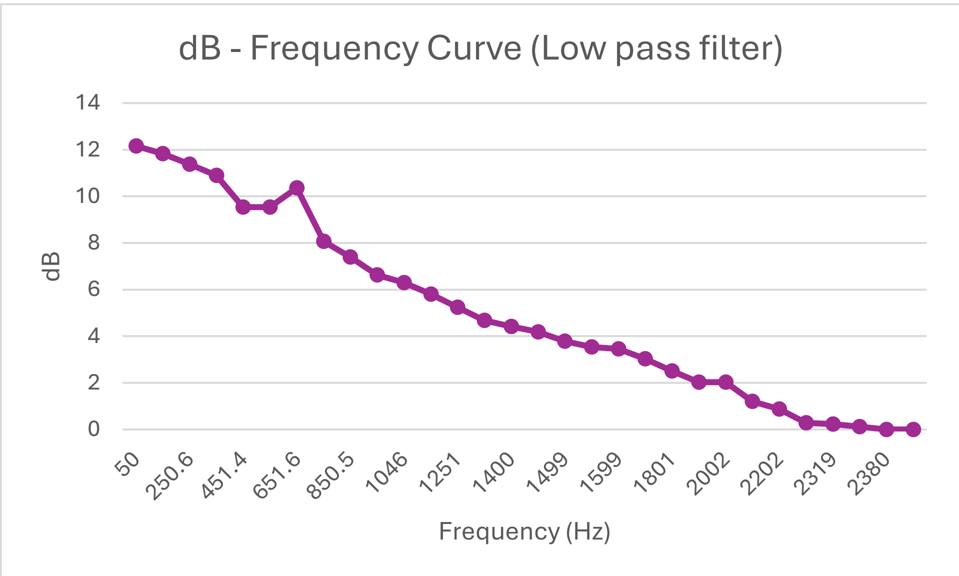

Figure 2. dB-Frequency Curve (Low pass filter)

(b) . Identify and report the two different bandwidths (with vs. without filter) of our OP-amp. Does the experimental cutoff frequency of low pass filter match its theoretical value? Estimate and report the error by :

$$Measurement\ error = \frac{f_{cutoff,exp} - f_{cutoff,theory}}{f_{cutoff,theory}} \times 100\%$$

We usually define the cutoff frequency as the frequency at which the output gain descends 3dB from the initial gain dB_0, the cutoff frequency is also known as the bandwidth.

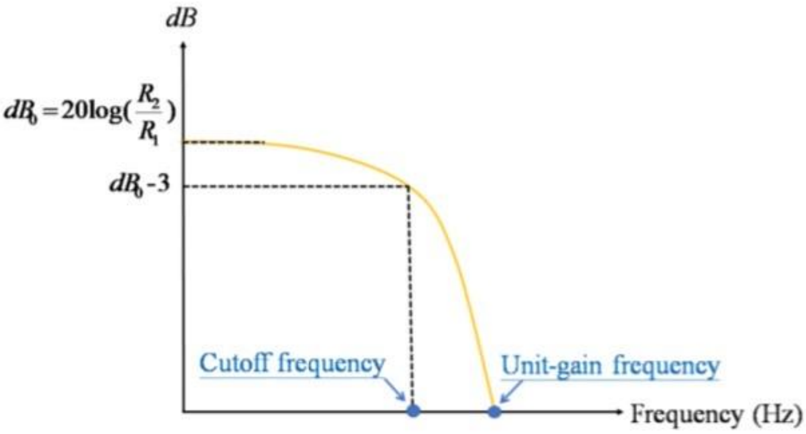

Figure 3. Decibel (dB) to frequency curve for inverting circuit (not a real scale)

A. For inverting circuit:

$dB_0 \approx 14.023$

$dB_0 - 3 = 11.023$

Table 3. dB-Frequency Chart for cutoff frequency (inverting circuit)

Frequency(Hz)	20log(Vpp,CH2 / Vpp,CH1)
50	14.02250633
20000	13.38600752
cutoff frequency	11.02250633
29970	10.91576049

By using interpolation, we can obtain that the cutoff frequency ≈ 29539.170Hz

B. For low pass filter:

$dB_0 \approx 12.157$

$dB_0 - 3 = 9.157$

Table 4. dB-Frequency Chart for cutoff frequency (low pass filter)

Frequency(Hz)	20log(Vpp,CH2 / Vpp,CH1)
50	12.15694523
550	9.542425343
cutoff frequency	9.15694523
750.7	8.068262986

By using interpolation, we can obtain that the cutoff frequency ≈ 602.481Hz

➔ theoretical cutoff frequency of low pass filter

$$= \frac{1}{2\pi Rc} = \frac{1}{2\times\pi\times500\times470\times10^{-9}} \approx 677.255\text{Hz}$$

$$Measurement\ error = \frac{602.481-677.255}{677.255}\times 100\% \approx -11.041\%$$